

Adaptive Robot Speed Control via Early-Exit CNN Inference Depth

Description: This project will investigate how early-exit convolutional neural networks (CNNs) can dynamically control a mobile robot's speed based on real-time inference complexity. Early-exit architectures attach intermediate classifiers to a CNN so that "easy" inputs exit early, reducing latency. This project wants to directly map the network inference speed to the robot's responsiveness.

For example, if the network exits early with high confidence (e.g., path is clear), the robot moves quickly. If the input requires deeper inference, indicating uncertainty or complexity, the robot slows down or stops.

The project will be evaluated on tasks such as obstacle avoidance or (more complex) object recognition/grasping. The idea is to extend an approach that has already been introduced with the confidence score in [1].

Keywords:

Early Exit Networks, Adaptive Inference, Mobile Robotics, Confidence-based Control, Dynamic Neural Networks

Resources:

[1] Speed Control of a Mobile Robot Using Confidence of an Image Recognition Model:

https://www.jstage.jst.go.jp/article/jrobomech/35/2/35_338/_pdf

[2] Analysis and Optimization of Early-Exits for Object-Detection:

<https://arxiv.org/pdf/2503.17497>

[3] Multi-Exit Vision Transformer for Dynamic Inference: <https://arxiv.org/abs/2106.15183>

[4] A Survey on Dynamic Neural Networks: from Computer Vision to Multi-modal Sensor Fusion: <https://arxiv.org/pdf/2501.07451v1>

Required level:

Master student with experience in computer vision and robotics.

Contact: fabmo@dtu.dk